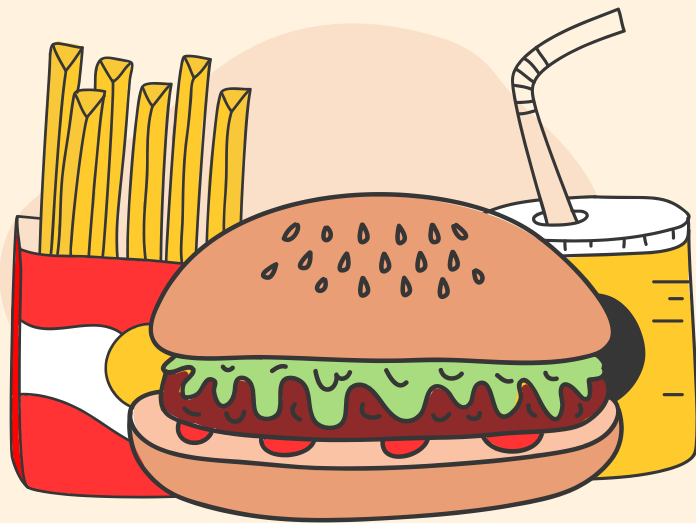
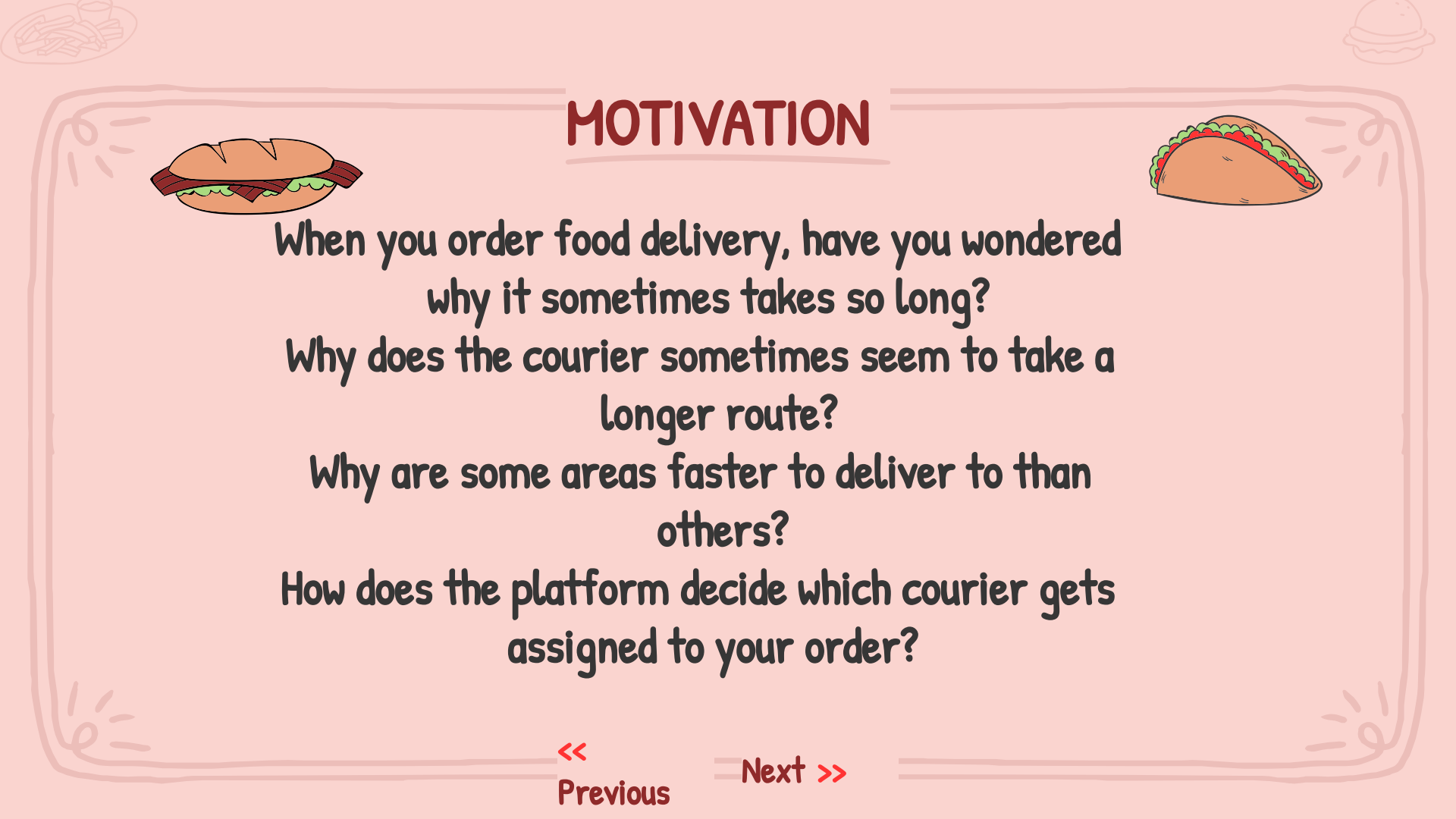




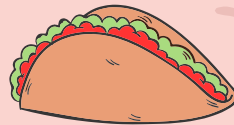
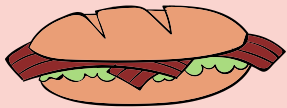
Courier Assignment and Bottleneck Detection in Urban Food Delivery Network

Bryce Gover & Tiangu Zhao

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MOTIVATION



When you order food delivery, have you wondered why it sometimes takes so long?

Why does the courier sometimes seem to take a longer route?

Why are some areas faster to deliver to than others?

How does the platform decide which courier gets assigned to your order?



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INTRODUCTION

- Food delivery is common in most cities, but the logistics behind it are largely invisible to the customer
- Two forces shape delivery efficiency: how orders get assigned to couriers, and what the road network itself allows
- Most existing research focuses on the assignment side. Less work treats the network itself as the binding constraint
- This project studies both, and how they interact



RESEARCH QUESTION



Structure: Where does the network break?

Bottleneck detection: Do simple metrics find the choke points, or is a more robust method needed?

Demand ceiling: Is there a volume the network simply can't move fast enough to meet?

Renovation targets: Which roads give the most improvement if upgraded?

Clustering effect: Do restaurant clusters help or hurt delivery?



Operations: Who delivers, and how?

Best strategy: which assignment method gives the fastest average delivery time?

Peak load: how does performance shift during lunch and dinner rushes?

Robustness: how much worse is a targeted road closure than a random one, and does strategy matter?

Fleet size: what's the ideal number of couriers for the network?



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Literature Reviews



A GRASP Algorithm for the Meal Delivery Routing Problem

Giraldo-Herrera &
Álvarez-Martínez
(2024)



Harvesting Efficient On-Demand Order Pooling from Skilled Couriers

Liang, Zhao, Li,
Feng, Zhang, Ding,
Hao & He (2024)



Real-Time Dispatching and Fleet Steering for Meal Delivery

Cheng & Azadeh
(2025)



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Literature Reviews



The Meal Delivery Routing Problem

Reyes, Erera,
Savelsbergh,
Sahasrabudhe &
O'Neil (2018)



Batching and Matching for Food Delivery in Dynamic Road Networks

Joshi, Singh, Ranu,
Bagchi, Karia & Kala
(2022)



Urban Amenity Agglomeration: Evidence from Milan Restaurants

Leonardi & Moretti
(2023)



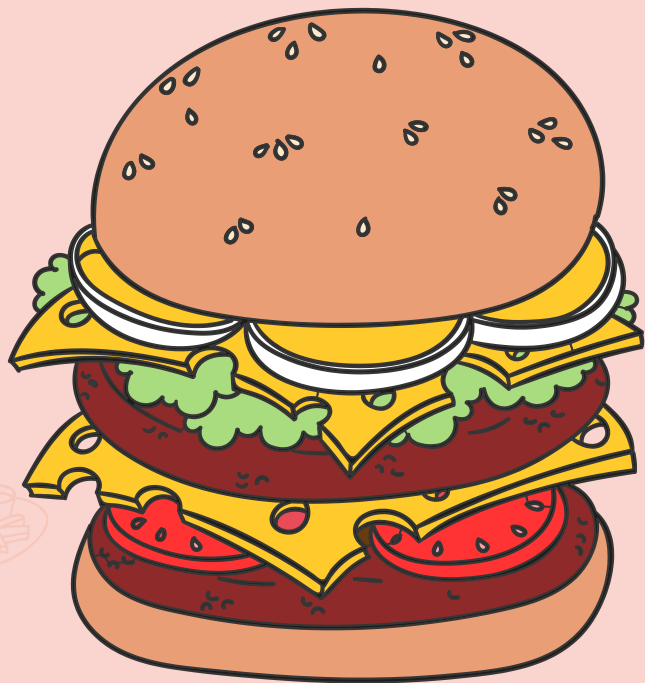
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METHODS

- **Road network structure:** OpenStreetMap street network data.
- **Restaurant locations:** Open POI data and local business licensing data.
- **Traffic and travel time:** Open traffic volume data and routing-engine travel-time estimates.
- **Population and demand proxy:** Census-style density data to simulate customer demand.



NETWORK COMPONENTS



Intersections

street intersections
from the road
network.



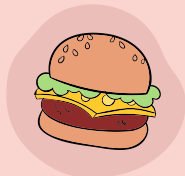
Restaurants

food pickup locations
from POI or business
data.



Delivery targets

simulated customer
drop-off locations.



Road edges and orders

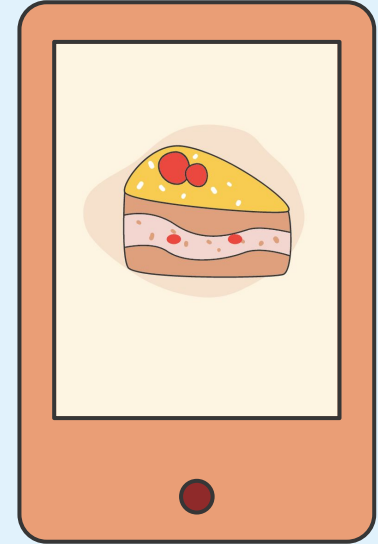
road segments are
weighted by travel time,
and orders are created
by pairing restaurants
with delivery targets.

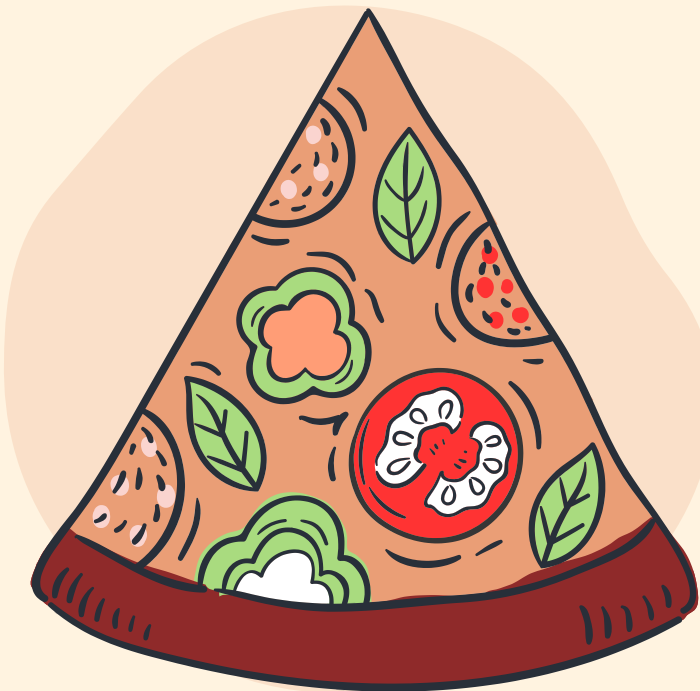
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THANK YOU For LISTENING!

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